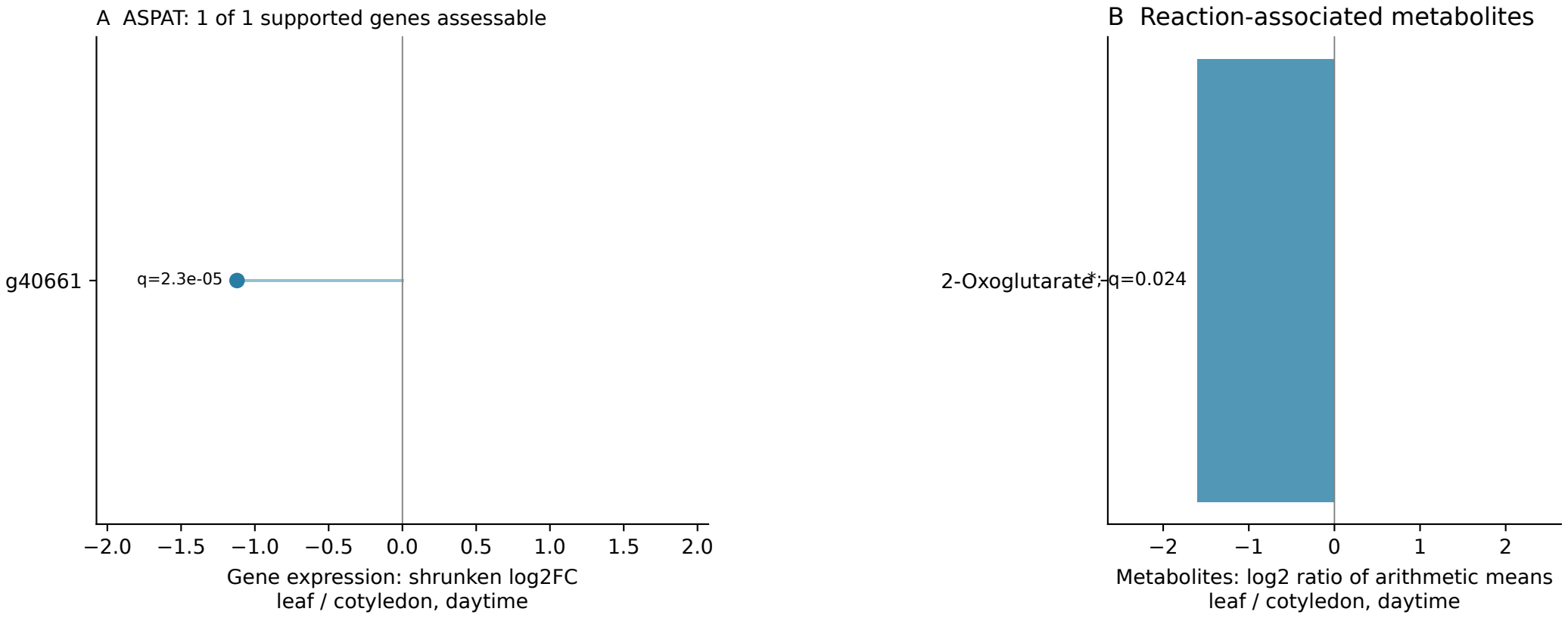



Unassessable genes omitted from plots; all genes retained in evidence tables. Filled dots: primary DE; open: not primary DE.
Metabolite *: BH q<0.05; ns: q≥0.05. Gene and metabolite q-values come from separate test sets.
Family-level reaction mapping does not establish gene-specific activity, flux or localization. Malate ratio may reflect a preprocessing floor.